

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MENTALCLOUDS | **Context:** Global South Mental Health Practitioners — India-Anchored OMHC / Peer-Support Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's expert evaluators are five Indian-affiliated healthcare professionals (two clinical psychologists, three psychiatrists/medical practitioners) [Q41, Q42], aged 40–55 with over a decade of experience [Q43], 4:1 male:female. Their institutional affiliations [Q9] (IIT Delhi, IIIT Delhi, AIIMS Rishikesh, AIIMS New Delhi, YourDOST Karnataka) provide partial alignment with the India-anchored deployment population, including one OMHC-affiliated organization (YourDOST). Inter-rater alignment is reported (variance lowest for MentalBART [Q50]; per-rater hallucination frequencies reported [Q52, Q56]; ratings averaged across five experts [Q47]). However, several gaps weaken convergent and external validity: (a) reference-summary creators' credentials and inter-annotator agreement at the data-creation stage are undocumented; (b) the senior-clinician profile (40–55, 10+ years) does not represent peer-support practitioners or younger OMHC counselors who form a meaningful share of the deployment population; (c) no Sub-Saharan African, Southeast Asian, or Latin American clinical practitioners were involved (annotator_representativeness gaps; [WEB-17, WEB-18]); (d) gender imbalance (4:1 male) may introduce blind spots around gendered disclosure patterns; (e) it is undocumented whether evaluators used CBT-specific, family-systems, or culturally adapted rubrics. OC was rated MODERATE, and the partial India alignment with explicit gaps justifies a middle score.

ID	Evidence
Q9	"Department of Electrical Engineering, Indian Institute of Technology Delhi, India; Department of Computer Science & Engineering, Indraprastha Institute of Information Technology Delhi, India; YourDOST, Karnataka, India; Department of Psychiatry, All India Institute of Medical Sciences, Rishikesh, India; Department of Psychiatry, All India Institute of Medical Sciences, New Delhi, India; Yardi School of Artificial Intelligence, Indian Institute of Technology Delhi, India" (p.1)
Q41	"In order to conduct a comprehensive expert assessment, five healthcare professionals were employed to assess the clinical appropriateness of the summaries produced by the LLMs based on the evaluation framework of Sekhon et al. [39]." (p.12)
Q42	"Among them were two clinical psychologists, with the remaining three comprising psychiatrists and medical practitioners." (p.12)
Q43	"Of the group, four were male, and one was female, with ages ranging from 40 to 55 years and possessing over a decade of therapeutic experience." (p.12)
Q47	"Table 6 reports the clinical experts' scores averaged over the ratings given by five experts." (p.13)
Q50	"Overall, all the raters were more aligned in rating the MentalBART model with lesser variance as compared to the other two LLMs for all the metrics." (p.13)
Q52	"Table 7. Hallucination frequency marked by experts for the top three LLMs. Here the average of hallucination frequencies for each rater are reported." (p.14)
Q56	"Table 7 presents the total numbers for each model and hallucination category as assessed by five individual raters." (p.14)
WEB-17	No non-Indian Global South annotator involvement found in published MentalCLOUDS paper (https://mental.jmir.org/2024/1/e57306)
WEB-18	No non-Indian Global South annotator involvement found in PMC-archived version (https://pmc.ncbi.nlm.nih.gov/articles/PMC11303879/)

Strengths

- Five expert raters with explicit reporting of per-rater hallucination frequencies [Q52, Q56] and acknowledgment that 'hallucinations are perceived [differently] among different models' [Q57] — supporting convergent-validity diagnostics.

- Indian institutional affiliation [Q9] including an OMHC platform (YourDOST) provides partial alignment with the India-anchored target population.
- Mix of clinical psychologists and psychiatrists [Q42] provides some disciplinary breadth within the senior-clinician profile.

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Q42	“Among them were two clinical psychologists, with the remaining three comprising psychiatrists and medical practitioners.” (p.12)
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Q57	“The data shows fluctuations in how hallucinations are perceived among different models and stresses the importance of reviewing evaluations from numerous appraisers for a complete assessment.” (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partial — ground-truth reference summaries' creators are undocumented (reference_annotation_creators flagged); expert acceptability ratings are by Indian senior clinicians [Q42, Q9] but it is unclear whether their judgments reflect peer-support/OMHC quality criteria or only senior-psychiatric standards.

OC-2: Likely disagreement on culturally salient content: Indian-context practitioners may rate summaries lower if family-system or spiritual coping content is omitted (A3); peer-support practitioners may apply different quality criteria than AIIMS-affiliated senior psychiatrists; non-Indian Global South practitioners are entirely unrepresented.

OC-3: Annotator demographics partially documented: five experts, 2:3 psychologist:psychiatrist split [Q42], 4:1 male:female, ages 40–55, 10+ years experience [Q43], Indian institutional affiliations [Q9]. No formal Datasheet/Data Statement disclosed in the paper. Reference-summary creator demographics undocumented.

OC-4: Re-annotation by a regional pool that includes peer-support/OMHC counselors and non-Indian Global South practitioners would meaningfully strengthen convergent validity; not done in source benchmark.

OC-5: Aggregation by averaging across five experts [Q47] could erase minority/divergent perspectives; the paper notes inter-rater variance, particularly higher for MentalLlama and Mistral [Q50], and stresses the importance of multiple appraisers [Q57], but does not separately analyze minority dissent.

OC-6: Documented harms: undocumented reference-summary creator credentials (potential convergent-validity gap); senior-clinician profile mismatch with peer-support deployment cohort; absence of non-Indian Global South annotators; potential aggregation-driven minority erasure.

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Information Gaps

- Reference-summary creator credentials, cultural backgrounds, and inter-annotator agreement at the data-creation stage.
- Whether evaluators used CBT-specific, family-systems, or culturally adapted rubrics when applying the Sekhon framework.
- Quality-criteria divergence between AIIMS senior clinicians and peer-support OMHC counselors.

Requires Expert Verification

- Direct comparison of how Indian peer-support counselors vs. senior psychiatrists rate the same model summaries.
- Inclusion of Sub-Saharan African, Southeast Asian, and Latin American clinical raters.

Remediation

Gap 1: Expert raters are exclusively senior India-based clinicians; peer-support practitioners and non-Indian Global South clinicians are unrepresented; reference-summary creator credentials undocumented.

Recommendation: Recruit a mixed annotator pool that adds peer-support/OMHC counselors, younger practitioners, and at least one Sub-Saharan African or Southeast Asian clinical practitioner. Document MEMO/MentalCLOUDS reference-summary creator credentials and inter-annotator agreement, and report minority-rating distributions rather than only averaged scores.